Collision lab

antocciphysics | TEACHER KEY | CPA

Learning target: Test momentum conservation with measured before-and-after velocities.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 4; 3 course days later.*
The new force magnitude is 1/3 of the original because the same momentum change occurs over 3 times the time.

02. Explain a worked example (worked)
Small decrease may involve friction/external impulse or measurement error; no cause is established by the difference alone.

03. Look and explain (model)
Both+.20 kg·m/s.

04. Build the solution (practice)
Total 2 kg·m/s before and after; consistent with momentum conservation.

05. Use the idea (reason)
Table must preserve raw data. Friction/track tilt; sensor noise or interval choice. Differences alone do not refute conservation.

06. Choose a method (mixed) *Review: Unit 2, Day 11; 12 course days later.*
4 s to top; 8 s total, assuming no drag and equal launch/landing heights.

07. Exit check (exit)
Opposite directions give opposite momentum contributions; unsigned addition misstates system momentum.

08. STRETCH: Apply it another way (stretch)
A interval -0.1 to 0.3 contains zero; B 0.6 to 1.0 does not.

Teaching note: Lab companion. Collect trials only with teacher-approved apparatus. Never label the supplied examples as your measured results.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 4; 3 course days later.*
J=-12 N·s; first F≈-12 N; longer-time F≈-6 N. Both oppose the initial velocity.

02. Explain a worked example (worked)
Small decrease may involve friction/external impulse or measurement error; no cause is established by the difference alone.

03. Represent and interpret (model)
Both+.20 kg·m/s.

04. Solve and justify (practice)
Before .300; after .2875 kg·m/s; deviation 4.17%.

05. Analyze the reasoning (reason)
Table must preserve raw data. Friction/track tilt; sensor noise or interval choice. Differences alone do not refute conservation.

06. Choose a method (mixed) *Review: Unit 2, Day 11; 12 course days later.*
Both ranges≈138.6 m. Times≈4 s and 6.928 s. The larger angle gives longer flight.

07. Exit check (exit)
5% deviation. Difference relative to initial is not an experimentally established uncertainty or probability.

08. STRETCH: Apply it another way (stretch)
Change from -0.4 to 0 kg·m/s; force -2 to 0 N. Bounds alone are not probabilities.

Teaching note: Lab companion. Collect trials only with teacher-approved apparatus. Never label the supplied examples as your measured results.